



Advanced Cisco Cybersecurity Academy – FortiGate-40F Factory Reset and Initial SOHO Configuration

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Purpose

This lab serves to develop a basic understanding and familiarity with Fortinet technologies and systems, and more specifically the FortiGate-40F. This lab provides the basic skillset for navigating the FortiGate-40F GUI and making basic SOHO (Small Office Home Office) level configurations. Understanding how to use Fortinet hardware, software, and systems serve as a valuable skillset as they hold 55% of unit market share in firewalls as of late 2025. This lab helps to develop those skills and a deeper understanding of concepts like network access and network security.

Background

Fortinet is a global leader in cybersecurity solutions, founded in 200 and headquartered in Sunnyvale, California. They occupy approximately 3.93% of the total market share within the entire IT field, and over half the market in firewall shipments. Because of these facts, HR and IT-decision makers value Fortinet certifications and skills highly, with 89% of leaders preferring to hire certified candidates to prove their skills with Fortinet Firewalls and systems. Fortinet has been able to be one of the top choices for IT solutions due to its AI-driven Security Fabric Architecture that integrates over 50 enterprise grade products to provide easier and enhanced central management systems, where less people can more easily and effectively manage larger and more complex architectures that incorporate multiple separate Fortinet products. Fortinet additionally is the only cybersecurity solutions company to use SPU's or Security Processing Units to give them an edge over their competition; these units work more efficiently and can handle higher workload when it comes to tasks related to cybersecurity compared to normal CPUs resulting in higher efficiency, performance, and low latency. These special advantages that Fortinet systems utilize have driven skills related to Fortinet to become increasingly valuable within the IT field.

The FortiGate-40F and all Fortinet devices come with the FortiOS. FortiOS is Fortinet's natively AI-powered and Quantum Safe operating system and serves as the foundation for converging networking and security to create a unified security platform. This approach allows Fortinet systems to eliminate any possible gaps in security created by individual devices and products by converging everything into one unified architecture. This way of managing security through a centralized security platform is a Security Fabric. Fortinet uses their Operating System to converge everything into a security fabric to then leverage AI to monitor the entire network, allowing for devices like the FortiGate-40F to provide high-level security to a network. This lab uses the FortiGate-40F to provide secure internet access to wired end devices in a SOHO environment. The FortiGate-40F can also be used to create a wireless network to access the internet without a physical connection but will be covered in a later lab. A SOHO environment is an environment where one to ten users are accommodated usually through one router/firewall, in this case a FortiGate-40F. This includes almost every household in the US and the millions of small businesses who all require a secure way to access the internet. Most of these households and businesses use ISP-supplied routers and gateways, and [Parks Associates](#) found that 71% of U.S. internet households lease, purchase, or receive routers from their ISP.

This sometimes is not a viable solution for businesses who require privacy and households who may include remote workers that deal with sensitive information, as ISP bought and rented routers can access all information that goes in and out of the network. This is where being able to configure a SOHO network using a firewall like the FortiGate-40F becomes a valuable skill to have. Additionally, being able to further isolate a SOHO network from an ISP's control allows for a significantly reduced chance of any sensitive information being leaked during attacks done on major ISPs. An example of SOHO networks getting attacked due to an ISP vulnerability is [an attack that compromised 600,000 SOHO routers within the span of three days](#) using a trojan to install remote access software using a firmware update as a guise. This attack only affected ISP issued routers, but isolating your device from an ISP does not mean the device cannot be comprised. [In 2021 61% of Small Businesses reported they received a cyberattack](#) this is where the skills of being able to configure a strong SOHO network become valuable to prevent data breaches that can lead to lawsuits and sensitive information being leaked.

To configure a strong network using the FortiGate-40F a factory reset is needed as this will reset and previous configuration or unknown configuration that could contain spyware or malware. Once the firewall is reset, the Firewall should allow for internet access with its default factory configurations. To create a strong network, access control policies and internet policies must be implemented. The access control policies control who is allowed to access the network and internet, this prevents anyone without the proper credentials from accessing the firewall's network from the inside of the network. Internet policies allow for control over what can be accessed on the internet, on the FortiGate-40F separate policies can be made on what can be searched, downloaded, and seen on websites. This allows for granular control on what can be done using the internet preventing intentional or accidental download of malware on end devices within the network.

Lab Summary

In this lab the FortiGate-40F is factory reset through using the factory reset button physically on the firewall rather than using the CLI. Admin credentials were configured along with a local user. Updates were configured to take place as often as possible and done automatically. Then as the FortiGate-40F Firewall allows for internet access with default configuration, only policy adjustments to increase security on the default outbound policy were made.

Lab Commands

To Factory Reset the Firewall the firewall must be plugged in and the user will have to be plugged into the firewall via console cable and let the firewall boot completely then have the Factory Reset button pressed within 60 seconds of booting.

```
FortiGate-40F (00:32-03.17.2023)
Ver:05000030
Serial number: FGT40FTK23099217
CPU: 1200MHz
Total RAM: 2 GB
Initializing boot device...
Initializing MAC... NP6XLITE#0
Please wait for OS to boot, or press any key to display configuration menu.....

Booting OS...
Initializing firewall...

System is starting...
Starting system maintenance...
Scanning /dev/mmcblk0p2... (100%)
Scanning /dev/mmcblk0p3... (100%)
FortiGate-40F login: admin

Password:
Verifying password...

System is resetting to factory default...
Login incorrect

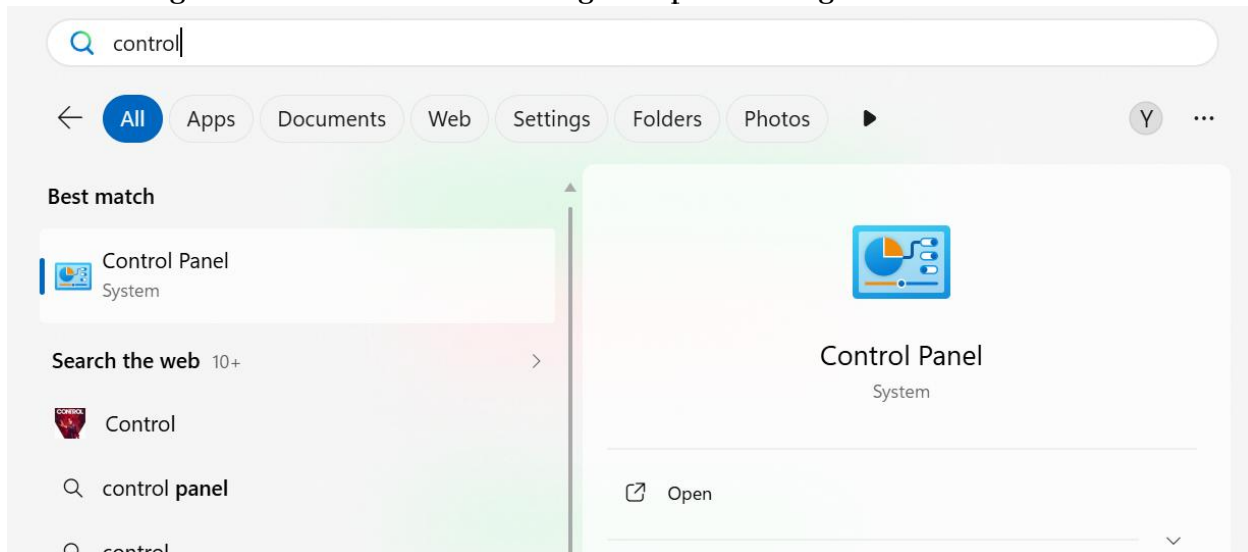
FortiGate-40F login:

FortiGate-40F login:

The system is going down NOW !!
```

To then access the Firewall via GUI the device connecting to the firewall being be connected to the firewall's LAN 1 port and have IP address and default gateway be automatically configured by the firewall.

Navigate to Control Panel > Change adapter settings > Ethernet

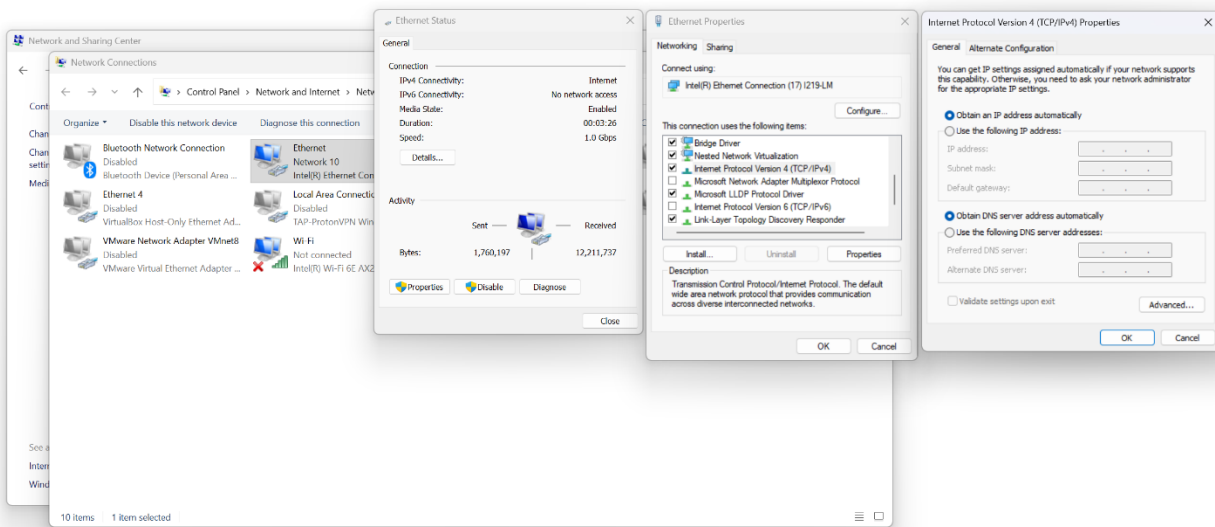


Control Panel Home

Change adapter settings

Change advanced sharing settings

Media streaming options

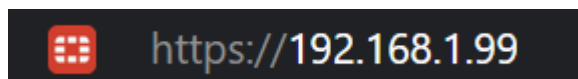


To ensure this is done correctly and the device connecting to the firewall has to proper IP addresses press the windows key and open command prompt by typing “cmd” and hitting enter. Then, type the command “ipconfig” and cross check your result under Ethernet Adapter Ethernet with the one shown below.

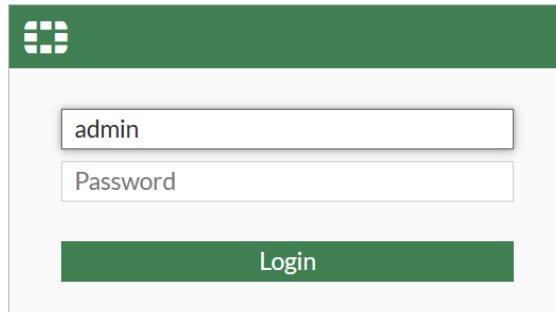
Ethernet adapter Ethernet:

```
Connection-specific DNS Suffix  . : 
IPv4 Address. . . . . : 192.168.1.110
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 192.168.1.99
```

If this is done correctly then in a web browser type <https://192.168.1.99> to access the GUI.

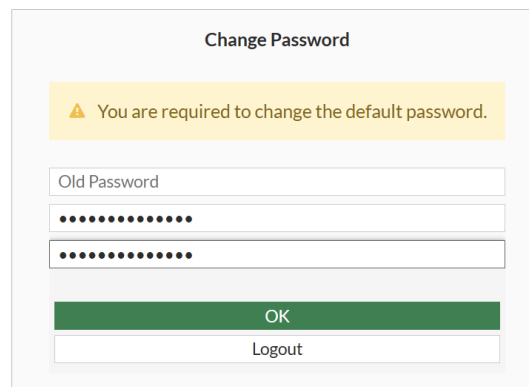


Then you will be prompted to enter a username and password. Ignore the password field and only enter “admin” as your username and login.

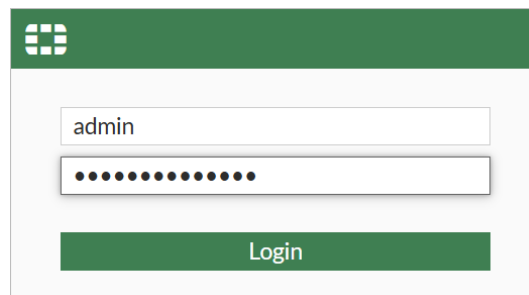


The image shows the FortiGate login interface. It features a green header with the FortiGate logo. Below the header, there is a white box containing two input fields: one for the username 'admin' and another for the password. A green 'Login' button is positioned below the password field.

You will then be prompted to enter a password and will need to re-login with the new password.

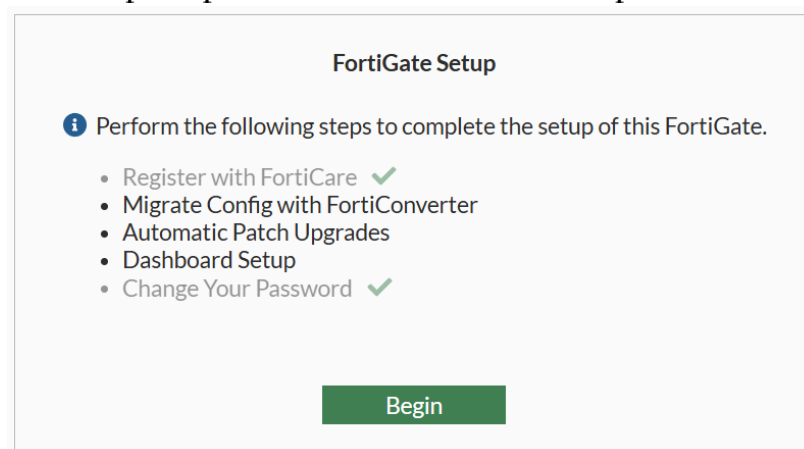


The image shows the 'Change Password' form. It has a title 'Change Password' and a yellow warning box that says 'You are required to change the default password.' Below this, there are two input fields for 'Old Password' and a new password, both masked with dots. At the bottom, there are two buttons: a green 'OK' button and a white 'Logout' button.



The image shows the FortiGate login interface again, identical to the first one. It features a green header with the FortiGate logo. Below the header, there is a white box containing two input fields: one for the username 'admin' and another for the password, masked with dots. A green 'Login' button is positioned below the password field.

You will then be prompted to do the FortiGate Setup



The image shows the 'FortiGate Setup' form. It has a title 'FortiGate Setup' and an information icon followed by the text 'Perform the following steps to complete the setup of this FortiGate.' Below this, there is a list of steps: 'Register with FortiCare' (checked), 'Migrate Config with FortiConverter', 'Automatic Patch Upgrades', 'Dashboard Setup', and 'Change Your Password' (checked). At the bottom, there is a green 'Begin' button.

Do not migrate from an older firmware unless needed.

Setup Progress
Register with FortiCare ✓
➤ Migrate Config with FortiConverter
Automatic Patch Upgrades
Dashboard Setup
Change Your Password ✓

Migrate Config with FortiConverter


Are you migrating from a different FortiGate with older firmware? The [FortiConverter Service](#) can convert the config so it is compatible with the current FortiOS version of this device. Conversions generally take one business day.

ConvertLater

Don't show again ☒

Configure the automatic patch upgrade to happen as often as possible

Setup Progress
Register with FortiCare ✓
Migrate Config with FortiConverter
➤ Automatic Patch Upgrades
Dashboard Setup
Change Your Password ✓

Automatic Patch Upgrades for v7.4

☒ **Enable automatic patch upgrades for v7.4**
Regularly check for available patch upgrades within the v7.4 version during a configured time window. If any are available, automatically download and install them. The device will reboot during the upgrade.

☐ **Disable automatic patch upgrades**
Do not automatically download or install patch upgrades.

Upgrade schedule Delay Specify days

Delay by number of days

Install during specified time to

PreviousSave and continue

Set up the dashboard to be comprehensive

Setup Progress
Register with FortiCare ✓
Migrate Config with FortiConverter
Automatic Patch Upgrades
➤ Dashboard Setup
Change Your Password ✓

Dashboard Setup

Select one of the following options to decide what dashboards will be available by default. You can always change your selection or manually customize your own dashboards later.

☐ **Optimal**
A set of popular default dashboards and FortiView monitors.

☒ **Comprehensive**
A set of default dashboards as well as all monitors and FortiViews. This set will be familiar to users coming from previous FortiOS versions

PreviousOKLater

Now you should be shown the dashboard, now navigate to the left navigation panel and open User and Authentication and go to User Definition and create a local user.

1 User Type

2 Login

Local User

Remote RADIUS User

Remote TACACS+ User

Remote LDAP User

FSSO

FortiNAC User

Create the login credentials

✓ User Type > **2 Login Credentials** > 3 Contact Info > 4 Extra Info

Username

Password 

Skip the contact info unless you wanted, in Extra Info create a user group named Admins for the user.

✓ User Type > ✓ Login Credentials > ✓ Contact Info > **4 Extra Info**

User Account Status ☒ Enabled ☐ Disabled

User Group ☒

Name

Type

Firewall

Fortinet Single Sign-On (FSSO)

RADIUS Single Sign-On (RSSO)

Guest

Members

✓ User Type > ✓ Login Credentials > ✓ Contact Info > **4 Extra Info**

User Account Status ☒ Enabled ☐ Disabled

User Group ☒

 Admins

Finish creating the User and then navigate back to the navigation panel and go to Network > Interfaces.

Under the Hardware switch on the list of interfaces choose lan and click edit.

+ Create New Edit Delete Integrate Interface Search			
	Name	Type	Members
802.3ad Aggregate 1			
fortilink	802.3ad Aggregate	a	
Hardware Switch 1			
lan	Hardware Switch	lan1 lan2 lan3	

Then under the network settings for the LAN interfaces copy the setting below.

Network

Device detection

Automatically authorize devices

STP

Security mode

Authentication portal

User access

Redirect after Captive Portal

Captive Portal

Local

External

Restricted to Groups

Allow all

Original Request

Specific URL

Network Exemptions

+ Create new

Edit

Delete

+ Search

ID

Source address

Destination address

Service

No results

Do the same to edit the wan interface.

+ Create New

Edit

Delete

Integrate Interface

Search

Name	Type
802.3ad Aggregate 1	
fortilink	802.3ad Aggregate
Hardware Switch 1	
lan	Hardware Switch
Physical Interface 1	
wan	Physical Interface

Edit to match the setting below, this will prevent anyone outside from connecting to the admin interface through the wan interface.

IPv4

HTTPS

FMG-Access

FTM

Speed Test

HTTP

SSH

RADIUS Accounting

PING

SNMP

Security Fabric Connection

Receive LLDP

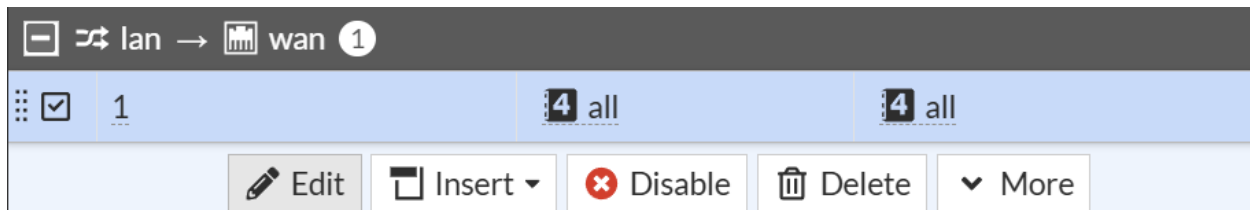
Use VDOM Setting

Enable

Disable

Confirm the configurations and navigate using the lefthand panel to Policy & Objects > Firewall Policy.

Select the policy named “1” that allows for traffic from the lan to go to the wan interface and be routed to the internet and select edit.



Rename the policy and copy the configurations below, this allows only the specified addresses to access the internet.

Name	Internet Policy
Incoming interface	lan
Outgoing interface	wan
Source	lan LAN_Ports address
Destination	all
Schedule	always
Service	ALL
Action	ACCEPT DENY

Under Firewall/Network Options ensure NAT is on and the outside interface is being used as outside IP addresses rather than a pool, to preserve the source port and keep the default protocol.

Firewall/Network Options	
NAT	☒
IP pool configuration	Use Outgoing Interface Address Use Dynamic IP Pool
Manage source port	☒ Fixed port Preserve source port
Protocol options	PROT default

Under Security Profiles, enable everything and select the default profile and enable SSL certificate inspection to make sure applied filters work.

Security Profiles	
AntiVirus	☒ AV default
Web filter	☒ WEB default
DNS filter	☒ DNS default
Application control	☒ APP default
IPS	☒ IPS default
SSL inspection	SSL certificate-inspection

Under Logging Options, log allowed traffic and enable to policy.

Logging Options

Log allowed traffic ☒ Security events ☐ All sessions

Comments

0/1023

Enable this policy ☒

Confirm the configurations and now update the firmware to the latest version. Update the firmware by navigating to System > Firmware and Registration in the left navigation panel.

Select your device and navigate to All Upgrades and select the newest one.

[Fabric upgrade](#) [Upgrade](#) [Register](#) [Refresh regist](#)

	Device
	FortiGate-40F

Select Firmware

Latest All Upgrades All Downgrades File Upload

7.6

7.6.6
Mature

FortiOS v7.6.6 build3652 (GA)
 [Release notes](#)
 [Security Fabric upgrade notes](#)

☐ Show More

The firewall will have to reboot, once rebooted log back into the firewall.

Now in the left Navigation panel under Setting go to FortiGuard to make sure all licenses are up to date and FortiGuard updates are set to be as often as possible.

FortiGuard Security Services

 Intrusion Prevention System (IPS) > ✔ Licensed (expires 2027/01/19)	 Advanced Malware Protection > ✔ Licensed (expires 2027/01/19)
 File analysis with FortiGate Cloud Sandbox query ⓘ > ✔ Licensed (expires 2027/01/19)	 AntiSpam query ✔ Licensed (expires 2027/01/19)
 URL, DNS & Video Filtering > ✔ Licensed (expires 2027/01/19)	

NOC & SOC

 FortiGate Cloud Log Retention ✔ Free License

FortiGuard updates

Scheduled updates



Every

Daily

Weekly

Automatic

Improve IPS quality ⓘ



Use extended IPS signature package



AntiVirus PUP/PUA ⓘ



Update server location ⓘ

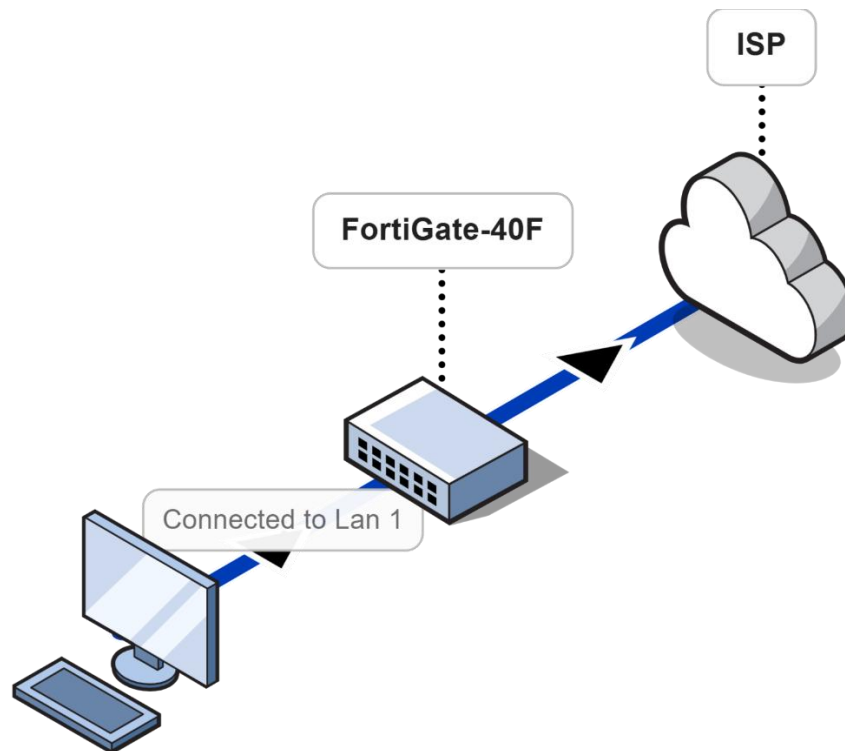
Lowest latency locations

US only

EU only

You have configured the FortiGate-40F with some basic security upgrades for a SOHO environment.

Network Diagram



Problems

The only problem regarding this lab I had was trying to Factory Reset the firewall, initially we tried to press the button before the firewall had completely booted up so the firewall would not reset. I then unplugged and re-plugged the firewall multiple times until I found out that I had to wait until the firewall had to completely boot. To know if the firewall was booted I consoled into the firewall and waited to press the button until the firewall had booted, even after knowing this it still took a couple tries before being able to factory reset the firewall.

Conclusion

This lab helped me gain an understanding of how Fortinet systems operate and their unique advantages that set them apart from their competitors. It also helped to develop an understanding of how security profiles and policies all converge together to create a secure connection to the internet. This lab was a very foundational and basic lab and the skills developed within the lab will greatly transfer to the next lab of creating a WLAN or Wireless Local Area Network that can be accessed from anywhere the firewalls signal can reach instead of cables.